CHEMISTRY ADAPTLY

USING RESOURCES – NPK FERTILISERS

KEY VOCABULARY

NPK fertiliser: supplies nitrogen, phosphorus and potassium.
Ammonia: from the Haber process — used to make nitrogen salts.
Neutralisation: acid + ammonia → ammonium salt fertiliser.
Phosphate rock: the source of phosphorus, treated with acids.

COMMON MISCONCEPTIONS

~~Fertilisers are made in one simple step.~~
✓ Industrial production has several stages.
~~Phosphate rock can be used directly.~~
✓ It is insoluble — it must be reacted with acid first.
~~NPK provides only nitrogen.~~
✓ It provides nitrogen, phosphorus AND potassium.

MAKING FERTILISER SALTS

Ammonia + nitric acid → ammonium nitrate.
Ammonia + sulfuric acid → ammonium sulfate.
Phosphate rock + acids → soluble phosphates.
Potassium from mined potassium chloride/sulfate.
NPK = nitrogen + phosphorus + potassium.

1 WHAT IS NPK

State the three elements supplied by an NPK fertiliser. **[3 marks]**

2 AMMONIUM SALT

Name the salt made from ammonia and nitric acid. **[1 mark]**

3 PHOSPHATE ROCK

Explain why phosphate rock is treated with acid before use. **[2 marks]**

4 LAB VS INDUSTRY

Suggest one difference between making a fertiliser salt in a lab and in industry. **[2 marks]**

5 FILL THE GAPS

Complete the sentence using the word bank. **[2 marks]**

_____ from the Haber process is neutralised by acids to supply _____.

nitrogen | potassium | ammonia | oxygen

6 WHY FERTILISE

Explain why farmers add NPK fertilisers to crops. **[1 mark]**

TIP: *NPK = Nitrogen, Phosphorus, Potassium — the key nutrients for plant growth.*

✓ Think: Why must phosphate rock be made soluble before plants can use it?